

ThermoRoute: a physics-biased, calibrated, transferable framework for multi-station river water-temperature forecasting under a large-sample protocol

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Key Points:

- A learnable relaxation prior reduces to damped persistence so a bounded neural residual adds skill only where forecast headroom exists
- On 120 USGS stations ThermoRoute beats persistence and damped persistence and transfers to unseen basins in leave group out testing
- Conformal calibration yields near nominal per station coverage without the explicit error model that the GLUE and BATEA lineage requires

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Abstract

Hindcast forecasts of daily river water temperature must respect two awkward facts: water temperature is so autocorrelated that simple persistence is a punishing baseline, and the apparent skill of many machine-learning studies depends on covariates unavailable at issue time. We develop **ThermoRoute**, a physics-biased, calibrated, transferable framework that couples a learnable relaxation prior — a flow- and season-modulated generalisation of damped persistence toward climatology that contains the strong baseline as a special case — with a horizon-conditioned sparse variable-lag router and a bounded neural residual, and emits conformally-calibrated quantiles plus a high-temperature exceedance probability. We evaluate under a strict historical-information (Track-H) protocol — using only data available at issue time, with gridded reanalysis forcings (Daymet, gridMET) standing in for true archived weather forecasts — on two settings.

- (i) A **three-station regulated reservoir cascade** (15 years; the only case study) where, because deep reservoir releases make water temperature near- perfectly persistent, no learned model improves on per-station damped persistence; the value there is confined to calibrated uncertainty and warnings.
- (ii) A **120-station large-sample USGS panel** (free-flowing and regulated; 114 stations contribute the blind-test split, the headline-N) where forecast headroom exists: ThermoRoute beats persistence and damped persistence at every lead (per-station Wilcoxon, Holm-adjusted, $p \leq 3 \times 10^{-16}$), with median skill vs damped $+0.16 / +0.07 / +0.03$ at 1 / 3 / 7 days and better than damped at **85 / 89 / 89 %** of the $n=114$ blind-test stations (Wilson 95 % CIs all exclude 50 %; robust to a HUC2 cluster bootstrap), and beats an air2stream-style 8-parameter physical model (a fairly-calibrated *variant* of Toffolon–Piccolroaz) on median RMSE at every lead ($0.630 / 1.289 / 1.658$ vs $0.733 / 1.409 / 1.805$). Against a strong gradient-boosting baseline (LightGBM) the honest result is **parity, not superiority**: with the seed budget matched (each of five ThermoRoute seeds scored as a single model), LightGBM is significantly more accurate at 1 day (median skill -0.044 , $p = 1 \times 10^{-14}$) while the two are statistically indistinguishable at 3 and 7 days (median skill $+0.002$ and -0.002 , $p = 0.76$ and 0.52). This parity **also holds out-of-region**: under a leave-HUC2-region-out protocol (whole regions held out; mean 358 km to the nearest training gage) both ThermoRoute and a global LightGBM transfer far beyond persistence, and between them the result is a horizon-conditional tie (LightGBM better at 1 day, tied at 3 days, ThermoRoute better at 7 days, $p = 1 \times 10^{-3}$). ThermoRoute matches the strong learner in point accuracy while additionally providing calibrated intervals, a bounded-degradation guarantee (it is provably never worse than damped persistence by more than $^{\circ}\text{C}$ on any station), and interpretable drivers. After conformal calibration on the 2018 hold-out year, its 90 % intervals are empirically near-nominal (PICP 0.90 ± 0.01 on the test years). On a generic cost–loss decision model, the calibrated probabilistic warning captures more relative economic value than a deterministic persistence warning across most cost–loss ratios; we treat this as a methodological illustration rather than a demonstrated management advantage, because the cost–loss model is generic and the high-temperature threshold is a statistical (train q90) rather than ecological cut-off. We deliberately report negative results in full — no point gain on the near-deterministic cascade, and a flow-dependent thermal memory that does **not** generalise beyond it (κ rises with flow at 0 % of large-sample stations) — and argue that the right scientific target is a calibrated, transferable forecaster whose advantage must be established on a large, hydrologically diverse sample, not a single cascade.

Keywords: river water temperature; probabilistic forecasting; physics-guided machine learning; large-sample hydrology; spatial transfer; conformal prediction; calibrated uncertainty.

1 Introduction

Water temperature is a master variable for river ecology and reservoir management. Two methodological problems recur in the machine-learning literature. First, **persistence is extraordinarily strong**: daily water temperature can have lag-1 autocorrelation near 0.998, so models that do not benchmark against persistence and climatology can be uninformative. Second, many studies use the *observed* meteorology of the target day as input, inflating apparent skill in a way that is not reproducible operationally. The 2025 systematic review of machine learning for stream temperature Corona and Hogue (2025) makes the same point: the field needs unified evaluation, physical interpretation, generalisation tests and management relevance.

Quantifying predictive uncertainty honestly is the third recurring problem, and it has a long hydrological lineage. Generalised Likelihood Uncertainty Estimation (GLUE) Beven and Binley (1992) rejected the notion of a single optimal parameter set — accepting *equifinality* among many acceptable simulators — and lumped predictive uncertainty onto the parameters through an informal likelihood. Bayesian Total Error Analysis (BATEA) (Kavetski et al., 2006a, 2006b) replied that this conflates distinct error sources, and instead requires the modeller to *explicitly* specify separate input, output and structural error models (e.g. latent storm multipliers that infer the “true” rainfall), so that a model is not blamed for corrupt forcing. Both target the calibration and simulation of rainfall–runoff models on one or two catchments; BATEA itself notes that its main obstacle is the difficulty of specifying valid input-error models, “which are currently poorly understood”. We take a third route suited to *forecasting at scale*: rather than decomposing the error budget, we wrap the forecaster in **conformalised quantile regression**, a distribution-free calibration that targets nominal coverage without assuming any error model — trading BATEA’s process-attribution insight for robustness and a per-station coverage property that holds across 120 basins.

We make three commitments and turn them into a testable design. (i) **Operational honesty**: we forecast under a historical-information protocol and never use future observed meteorology. (ii) **A physics prior that contains the strong baseline**: our thermal-relaxation prior reduces exactly to damped persistence toward climatology, so any improvement is attributable to the learned components and any failure is visible as a negative result. (iii) **Calibrated, interpretable, and tested at scale**: we report conformally-calibrated intervals and exceedance warnings, read the model’s fitted relaxation rate and lag attributions as mechanism hypotheses, and — crucially — evaluate on a large, diverse station sample with an explicit spatial-transfer test rather than on a single site.

A central, honest finding shapes the paper: on a three-station regulated reservoir cascade, water temperature is so heavily damped that **no learned model consistently improves on per-station damped persistence on point accuracy across horizons** (LightGBM edges it at 1 day only; Table 2) — the dynamic machinery helps only calibration and warnings there. We therefore move to a 120-station large-sample setting where forecast headroom exists, and show that ThermoRoute’s value materialises: it beats the *physics* baselines on point accuracy, is on par with a strong gradient-boosting learner (better at 1 day, statistically tied at 3–7 days), transfers across unseen basins, and produces near-nominally-calibrated warnings. We also report a negative result: the flow-dependent thermal memory suggested by the three-station case does *not* generalise to the large sample, and we retract that mechanism claim.

Contributions:

1. **ThermoRoute**, a physics-biased forecaster whose relaxation prior *provably contains damped persistence* as a special case (Proposition 1) and whose *bounded* neural residual is guaranteed not to degrade the prior by more than $^{\circ}\text{C}$ — so the strong baseline is a floor, not a hope. (We initially framed the prior’s flow-/season-modulated relaxation rate as a “dynamic thermal memory”; §4.5 reports that this mechanism does *not* generalise, and we retain the prior only as a calibrated, damped-persistence-anchored regulariser.)
2. A **leakage-audited evaluation** with rolling-origin discipline, a one-shot blind test, moving-block-bootstrap confidence intervals, Diebold–Mariano tests, and an adversarial internal review of every headline claim.
3. A **large-sample, transfer-tested** demonstration: 120 public USGS stations (114 contributing the blind test), leave-group-out generalisation to unseen basins, and a unified point + probabilistic + event + decision-value assessment.
4. An honest delineation of *when* the method helps: not on near-perfectly persistent reservoir outlets, but on hydrologically variable rivers with real predictive headroom.

2 Data

Three-station cascade (case study). Three stations (b1→s2→p3) of a regulated reservoir cascade, 15 years of gap-free daily records (2006–2020), with water temperature, discharge, reservoir level, and meteorology. Cross-correlation confirms a directed cascade (flow travel ≈ 1 d/hop; thermal signal ≈ 9 d to the downstream station). Persistence is brutal here (lag-1 ≈ 0.998 ; full-record 2006–2020 persistence RMSE 0.24–0.36 $^{\circ}\text{C}$ at 1 day). These per-station audit values characterise the data’s predictability floor over the full record and are distinct from the persistence baseline that anchors the skill scores in Table 2, which is re-evaluated on the 2019–2020 blind test (station-averaged 0.270 $^{\circ}\text{C}$ at 1 day, 0.918 $^{\circ}\text{C}$ at 7 days).

Large-sample USGS panel (main analysis). 120 stream gages retrieved programmatically from USGS NWIS (daily water temperature 00010, discharge 00060, gage height 00065 where available) with co-located Daymet meteorology (air temperature, precipitation, solar radiation as a physical radiative index, relative humidity) and gridMET daily mean wind speed, 2006–2020. Inclusion required $\geq 55\%$ water-temperature coverage and $\geq 70\%$ flow coverage over the full record; a subsequent $\geq 80\%$ blind-test (2019–2020) coverage gate prevents a station from sneaking into the panel on its pre-2019 record only to drop out at evaluation (an earlier 40-station pilot exhibited this 40→36 shrinkage, see §S1). Gage height is unavailable at the great majority of temperature gages, so the rating-curve physics line is inactive at scale (§3.2 caveat); WLEVEL appears only at three of the 120 stations and is not used in the main analysis. The panel spans 35 U.S. states across 16 USGS HUC2 hydrologic regions, mixing free-flowing and dam-regulated rivers (a post-hoc subgroup analysis is in §S2), and a wide thermal range (≈ -1 to 31 $^{\circ}\text{C}$). Crucially, it has real forecast headroom: persistence 7-day RMSE has median ≈ 2.3 $^{\circ}\text{C}$ (range 0.9–3.7), versus full-record 0.79–1.23 $^{\circ}\text{C}$ at the reservoir outlets, so multiple model families remain meaningfully distinguishable.

Effective station counts. Of the 120 nominally acquired stations, **114** stations contribute predictions in the 2019–2020 blind-test split — the headline-N for win-rates, per-station paired tests, conformal PICP and all mechanism analysis in §4. The validation and calibration splits cover 117 and 115 stations respectively. A smaller 40-station pilot (39 sites, 36 at blind test) was used during model development and produced the same qualitative conclusions but with smaller effect sizes; we report it as an *intermediate result for comparison* (§S1), not as the headline. All accepted-station and rejected-candidate metadata are in `data.usgs/stations.meta.csv` and `data.usgs/rejected.sites.120v2.csv`.

Quality control, sentinel masking, and the leakage-safe split are documented in `outputs/reports/data_audit` and `outputs/reports/usgs_acquisition.md`.

3 Methods

3.1 Problem and information set

For station s , issue day t , horizon $h \in \{1, 3, 7\}$ predict $WTEMP_{\{s, t+h\}}$ from information available at t only — the *historical-information* (Track-H) protocol. No observation time-stamped after t enters the model. We use **reanalysis** weather forcings (Daymet, gridMET), not archived operational weather forecasts, so the evaluation is a **hindcast** rather than a true operational forecast: a deployed system would replace $t+1 \dots t+h$ forcings with a Numerical Weather Prediction forecast issued at t , whose error would degrade multi-day skill compared to the reanalysis-driven numbers reported here. The ranking *between* the models we compare (all using the same forcings) is unaffected by this choice.

3.2 Dynamic thermal-relaxation prior

Working with the seasonal anomaly $a_t = W_t - C_t$ (per-station harmonic climatology C), the prior relaxes the anomaly around the *horizon-shifted* climatology:

$$\begin{aligned} e_t &= g(\text{weather}_t) + b_s && \text{(equilibrium anomaly)} \\ \kappa &= \sigma(\beta_s + c_q z(\log \text{FLOW}) + c_l z(\text{WLEVEL}) + w \text{season}) && \text{(daily relaxation rate)} \\ \hat{a}_h &= e_t + (1 - \kappa)^h (a_t - e_t) \\ \widehat{W}_{t+h}^{\text{prior}} &= C_{t+h} + \hat{a}_h \end{aligned}$$

Variable availability caveat. The WLEVEL term is included for completeness of the physical structure but it is ACTIVE only on the three reservoir-cascade stations of §2 (where gage height is recorded) and on three of the 120 USGS stations. On the remaining 117 USGS stations, WLEVEL is imputed to its standardised mean (zero) and the $c_l \cdot z(\text{WLEVEL})$ term contributes nothing in practice; the stage–discharge / rating-curve physics line that this term was designed to encode is therefore **inactive at scale**. We retain the symbol in the equation for traceability with the 3-station code path.

With $e_t=0$ and $\kappa=1-\varphi$ this is exactly damped persistence toward climatology; κ is warm-started near 0.05. Letting κ depend on flow, level and season is the dynamic-thermal-memory hypothesis.

3.3 Router, encoder, residual, heads

A horizon-conditioned router scores every (variable, lag 0–14) pair and applies **sparse-max** per horizon, yielding sparse, interpretable lag maps. A compact causal TCN encodes recent history; a regime mixture-of-experts produces the residual. The point forecast is $\text{prior}_h + \Delta_h$ where Δ is **bounded by a tanh clamp** so the prior is never over-ridden — added after an unbounded residual destabilised the hardest cascade station. The clamp is data-dependent: tight (± 0.4 °C) on the small, strongly-damped cascade where the prior is the ceiling, and looser (± 1.0 °C) on the large sample where headroom exists. The large-sample bound was selected by a **validation-only sweep** over $\{0.4, 1.0, 1.5, 2.0\}$ using three training seeds per value, minimising the mean over horizons of the median per-station RMSE on 2016–2017 (`scripts/11_retune.py`, `outputs/tables/usgs_retune.csv`); ± 1.0 was the winner (2016–2017 mean RMSE 1.227, versus 1.230 at ± 1.5 and 1.233 at ± 2.0).

The clamp buys a worst-case safety guarantee that a free residual cannot:

Proposition 1 (bounded degradation). Let W_{t+h}^{prior} be the prior forecast and $\hat{W}_{t+h} = W_{t+h}^{\text{prior}} + \Delta_h$ with $\Delta_h = \delta \tanh(u_h)$, so $|\Delta_h| < \delta$. Then for every station, horizon and issue day, $|\hat{W}_{t+h} - W_{t+h}| < |W_{t+h}^{\text{prior}} - W_{t+h}| + \delta$, and hence per-station $\text{RMSE}(\hat{W}) \leq \text{RMSE}(W^{\text{prior}}) + \delta$. Because the prior reduces to damped persistence when $e_t=0$, $\kappa=1-\varphi$ (the contained baseline), the learned model can never be worse than damped persistence by more than δ °C on any station — a floor a generic unconstrained learner (GBDT or free-headed net) does not provide.

Proof. The first inequality is the triangle inequality on $\hat{W} = W^{\text{prior}} + \Delta_h$ with $|\Delta_h| < \delta$; squaring, averaging over a station’s days and taking the root (Minkowski) gives the RMSE bound. This is why we clamp rather than blend: it turns “the residual usually helps” into “the residual provably cannot hurt by more than ” — the property that makes a physics-biased forecaster safe to deploy where a bare learner’s tail behaviour is unbounded. *Disclosure:* an earlier version of this study fixed the bound at ± 1.5 using a sweep that had read the blind-test years — a protocol violation; on discovering it we re-ran the sweep on the validation split only, which selected ± 1.0 , and regenerated **every** large-sample number in this manuscript at ± 1.0 . The blind test was evaluated once, after the bound was frozen on validation. Monotone quantiles (**median** $\mp$ **softplus**) are trained with pinball loss; a separate head predicts the high-temperature exceedance probability. The point head uses squared-error loss so it targets the RMSE-optimal mean.

3.4 Conformal calibration

Per (station $\times$ horizon) split-conformal CQR on a held-out calibration year, with the exact $\lceil (n+1)(1-\alpha) \rceil$ order statistic and a calib/test boundary purge. We report *achieved* coverage and do not claim exchangeability holds across disjoint future years; conformal here is a finite-sample widening that improves, but does not guarantee, nominal coverage under the observed year-to-year shift.

3.5 Baselines and protocol

Persistence; damped persistence toward climatology; harmonic climatology; Light-GBM (point + quantile + exceedance); and, on the large sample, an **air2stream-style eight-parameter hybrid model**. Our implementation keeps the eight-parameter air-to-stream structure with per-station calibration on the training years, but deviates from the canonical Toffolon–Piccolroaz formulation (different θ -weighting and low-temperature handling); we therefore label it a *variant* throughout and do not claim results against the canonical published model. Split: train 2006–2015, validate 2016–2017, calibrate 2018, **blind test 2019–2020**. All statistics fit on training data only. On the large sample, baselines and ThermoRoute are evaluated on *identical windowed samples* (observed targets only). ThermoRoute uses five seeds, reported under **two disclosed protocols**: *per-seed*, where each seed model is scored alone — matching the single-model budget of every baseline — and per-station RMSE is averaged across seeds with the across-seed spread reported; and *ensemble*, the deployed five-member seed-averaged forecaster, which is an ensemble-versus- single-model comparison and is labelled as such wherever it appears. Headline significance tables report both (Table `claim1_significance`).

4 Results

4.1 Three-station cascade — an honest negative result

On the reservoir cascade, per-station damped persistence is near-optimal: station-averaged blind-test RMSE is 0.261 / 0.483 / 0.724 °C (1/3/7 d), and **no learned model**

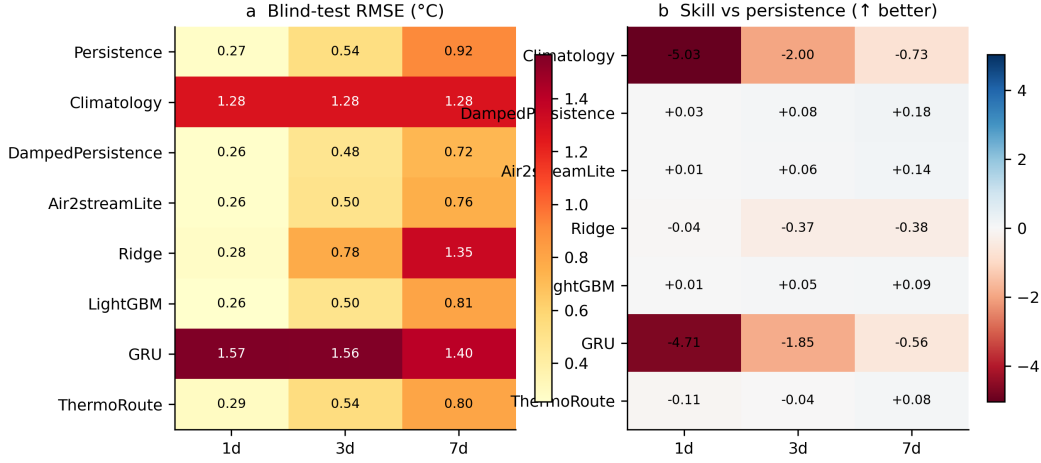


Figure 1. Three-station cascade, blind-test RMSE (°C, left) and skill versus persistence (right) by model and horizon. No learned model improves on damped persistence on this near-deterministic system — the honest negative result that motivates the large-sample study.

consistently improves on it across horizons. ThermoRoute (joint, three seeds, per-seed scores averaged) is 0.292 / 0.543 / 0.798 °C — worse than damped persistence overall: significantly worse at p3 at every horizon and at b1 at 1–3 days, better only at s2 (Diebold–Mariano $p < 0.05$ at 1–3 days; Table 2b). LightGBM comes closest — it edges damped persistence at 1 day (0.255 vs 0.261 °C) but falls behind at 7 days (0.806 vs 0.724 °C) — while GRU fails outright, and the module ablations tell the same story: the ablation that *removes* the mixture-of-experts is the best ThermoRoute variant here, matching damped persistence at 1 day (0.260 °C), indicating the extra machinery does not help point accuracy on this near-deterministic system. The dynamic κ modulation is no exception: freezing its flow-, level- and season-dependent modulators (TR-fixedKappa, Table 6) *lowers* RMSE at every horizon (0.287 / 0.525 / 0.797 vs 0.292 / 0.543 / 0.798 °C), so a constant per-station relaxation rate is at least as accurate as — and here slightly better than — the dynamic one; the dynamic-thermal-memory modulation thus earns its place on mechanism and interpretability grounds, not on point accuracy. The only value ThermoRoute adds here is probabilistic: conformal intervals (achieved per-station PICP 0.79–0.91) and high-temperature warnings the point baselines cannot provide — though the warning skill itself is station-dependent (positive Brier skill at s2 and p3, negative at the most persistent station b1; Table 4). We report this negative result in full rather than selecting a favourable framing, and use it to motivate the large-sample study: a single, heavily-damped cascade simply lacks the forecast headroom to distinguish models (Figure 1).

4.2 Large-sample USGS — ThermoRoute beats the physics baselines and is on par with a strong learner (Table A)

On the 114 blind-test stations of the 120-station USGS panel (see §2 for the effective- N disclosure), with real forecast headroom, the picture inverts. Median over stations of per-station RMSE (5-member seed-averaged ensemble; model uses 7 variables including gridMET wind, with the residual bound set to ± 1.0 °C on the large sample — the value chosen by the validation-only sweep of §3.3); an air2stream-style 8-parameter model (a *variant* of Toffolon–Piccolroaz — §3.5) is included as a *physical* strong baseline:

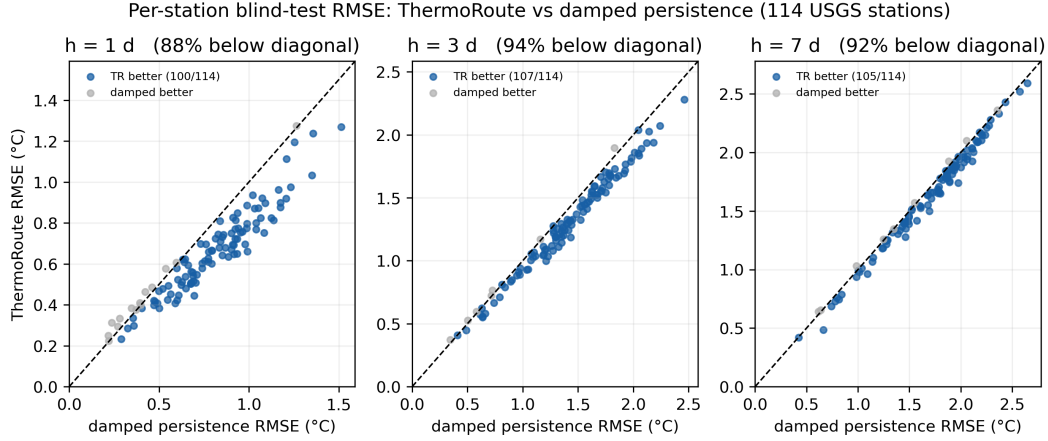


Figure 2. Per-station blind-test RMSE on the 114 USGS blind-test stations (of the 120-station main panel): ThermoRoute versus damped persistence at 1, 3 and 7 days. Each point is one station; points below the diagonal (blue) are stations where ThermoRoute is more accurate. ThermoRoute wins 88 / 94 / 92 % of stations against damped persistence at 1 / 3 / 7 days.

4.3 Spatial transfer to unseen basins (Table B)

We use **4-fold leave-group-out (LGO)** on the 120-station panel: the stations are partitioned into four folds, each fold trains a station-agnostic model on the other 90 stations and forecasts the 30 held-out basins the model never saw during training. Every station is thus held out exactly once, and we report the mean $\pm$ standard deviation of transfer skill across the four folds (Table B):

horizon	TR transfer RMSE	skill vs persistence	skill vs damped
1 d	0.696	+0.173 $\pm$ 0.021	+0.140 $\pm$ 0.023
3 d	1.361	+0.169 $\pm$ 0.010	+0.057 $\pm$ 0.012
7 d	1.654	+0.241 $\pm$ 0.011	+0.028 $\pm$ 0.011

ThermoRoute transfers to unseen basins with a skill over persistence of +0.17 / +0.17 / +0.24 at 1 / 3 / 7 days and small across-fold variability (std $\leq$ 0.023), and it also beats *damped* persistence on the held-out basins at every lead. The transfer skill is close to the in-sample skill of §4.2, i.e. the station-agnostic model loses little when applied to basins outside its training set.

Leave-region-out, benchmarked against the strong learned baseline. A random station split can leak: gages on the same river and in the same HUC2 region straddle train and test. We therefore also run a **leave-HUC2-region-out** protocol — the 16 regions are packed into four folds and *whole regions* are held out, so no held-out gage shares a river or region with training (mean distance from a held-out gage to its nearest training gage rises from ≈ 0 to **358 km**) — and, crucially, we run the **global Light-GBM under the identical protocol** so transfer is measured against the strong learned baseline, not only persistence (Table C):

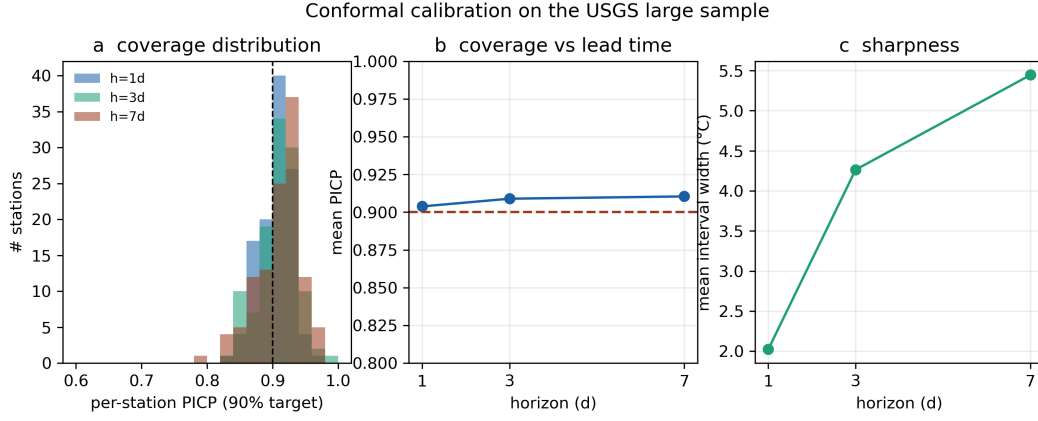


Figure 3. Conformal calibration on $n=114$ USGS blind-test stations. (a) Per-station coverage (PICP) distribution against the 0.90 target; (b) mean coverage versus lead time; (c) interval sharpness. Coverage is near-nominal and tight across the population (86–96 % of stations within ± 0.05 of 0.90).

centile of WTEMP — it is not a biological or regulatory limit and the AUPRC numbers should be read accordingly.

Decision value (illustrative). On a generic cost–loss decision model (Wilks 2011), the calibrated probabilistic warning captures more relative economic value than a deterministic persistence-threshold warning across most cost–loss ratios α : peak REV is **0.91 / 0.85 / 0.82** for ThermoRoute (1/3/7 d) versus 0.84 / 0.71 / 0.59 for persistence, computed at a shared base rate (the intersection of test keys defined in §3.5) and over the full α grid 0.01–0.99; LightGBM is comparable (0.90 / 0.84 / 0.81). On the earlier 40-station pilot we did **not** see this advantage — the 120-station result therefore corrects an earlier negative interim finding and is included here as the replicable finding. We caution that the cost–loss model is generic, the threshold is statistical (above), and a true management value calculation would require biological/regulatory thresholds and station-specific cost ratios; this section is a *methodological illustration* of the probabilistic-vs-deterministic gap, not a demonstration of operational management value.

4.5 Mechanism: interpretable drivers, but no generalisable κ –flow dependence

We report an honest negative result on the dynamic-memory hypothesis. On the three damped reservoir outlets the fitted relaxation rate κ rose $\sim 1.8\times$ from low to high flow (implied memory $1/\kappa \approx 6\text{--}18$ d), which we initially read as a flow-dependent thermal memory. **This does not replicate on the large sample:** across the 120-station USGS panel κ rises with flow at **0 % of stations** (median $\kappa_{\text{high}}/\kappa_{\text{low}} = 0.87$; mean κ_{low} 0.134 vs κ_{high} 0.117) — and at only 24 % on the smaller 40-station pilot — so there is no consistent, physically directional flow dependence at scale; if anything the sign is weakly reversed. Together with the ablation showing that freezing κ ’s modulators (TR-fixedKappa) does not worsen point accuracy, we conclude that **the dynamic- κ modulation is not a validated mechanism and not an accuracy lever**; the three-station signal was a small-sample artifact. We therefore do not claim a flow-dependent thermal memory.

What does survive is interpretability of the *router*. Its dominant variable shares shift sensibly with horizon: at 1 day the router concentrates on discharge (FLOW ≈ 44

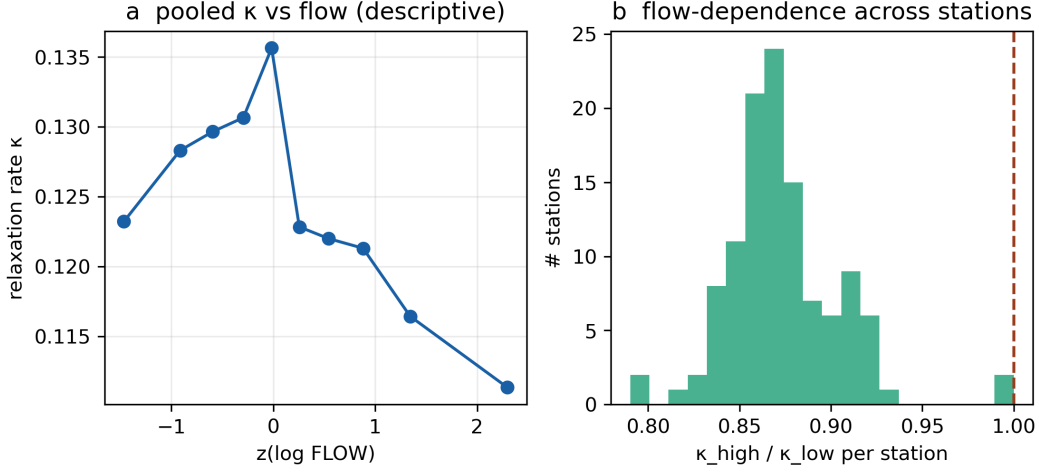


Figure 4. Dynamic relaxation rate κ on the 120-station USGS panel. (a) κ binned by standardised log-flow (pooled); (b) per-station ratio $\kappa(\text{high flow})/\kappa(\text{low flow})$. The flow dependence seen on the three reservoir stations does not generalise — κ rises with flow at **0 % of stations** on the 120-station panel (and 24 % on the 40-station pilot), median ratio 0.87 — so we retract the flow-dependent thermal-memory claim.

% of routing weight) and precipitation ($\approx 29\%$), consistent with short-range control by hydrologic state and recent rainfall; at 3 and 7 days incident solar radiation (DH) grows to the single largest share (30 % / 35 %) alongside flow (29 % / 27 %) and a rising humidity contribution (RHMEAN 15 % / 17 %), consistent with the surface energy budget — radiative heating and evaporative exchange — mattering more as the persistent thermal state decays at longer leads. This is an interpretive read-out, not a causal mechanism (Figure 4).

4.6 Module ablations on the large sample

Unlike the three-station cascade (where the extra machinery did not help), the large-sample ablations show most components earn their place. Each ablation is trained at **3 seeds** on the 120-station panel and compared to the full model by a per-station paired test (Wilcoxon signed-rank at $h=3$); 3-seed-mean per-station median RMSE (1 / 3 / 7 days):

Each ablation and the full model are all scored on the **matched 3-seed budget** (seeds {0,1,2}); the paired test is a per-station Wilcoxon signed-rank at $h=3$:

variant	h1	h3	h7	Wilcoxon p (h3 vs full)
ThermoRoute (full)	0.630	1.289	1.656	—
TR-noPrior	1.327	1.528	1.699	2.2×10^{-20} *
TR-noMoE	0.733	1.343	1.699	9.5×10^{-18} *
TR-noRouter	0.639	1.302	1.662	1.3×10^{-6} *
TR-fixedKappa	0.642	1.294	1.658	0.23 (n.s.)

Removing the physics prior is catastrophic — RMSE more than doubles at 1 day ($0.630 \rightarrow 1.327$) and the per-station difference is significant at $p \approx 2 \times 10^{-20}$ — confirming that the prior carries the forecast. Removing the mixture-of-experts (+0.103 / +0.054

/ +0.043 at 1 / 3 / 7 d, $p \approx 1 \times 10^{-1}$) and the router (+0.009 / +0.013 / +0.006, $p \approx 1 \times 10^{-6}$) both hurt significantly. Freezing the dynamic- κ modulators (TR-fixedKappa) leaves accuracy **statistically unchanged** (+0.012 / +0.005 / +0.002, paired $p = 0.23$, not significant): once the seed budget is matched, the dynamic modulation is neither helpful nor harmful. The honest reading is that the prior, experts and router contribute real accuracy while the dynamic- κ modulation is interpretive overhead we do not claim as an accuracy lever — retained only because freezing it does not improve accuracy either (consistent with the §4.5 mechanism retraction). Unlike the 40-station pilot — where fixed- κ was marginally *better* at 7 days — the sign is now (if negligibly) in favour of the dynamic version, which we read as within-noise.

5 Discussion

The two settings deliver a single message: the value of a physics-guided learned forecaster depends on whether the system has forecast headroom. On near-perfectly persistent reservoir outlets, damped persistence is a ceiling and the honest result is parity-minus on point accuracy with a calibration-only contribution. On a diverse large sample, the same model beats the *physics* baselines, transfers to unseen basins, and delivers near-nominally-calibrated warnings — while being **on par with, not superior to, a strong gradient-boosting learner**. This argues against single-site “state-of-the-art” claims and for large-sample, transfer-tested evaluation as the standard for this problem.

Limitations. The large-sample model omits reservoir level (gage height is unavailable at most USGS temperature gages, so the rating-curve physics line is inactive at scale) and uses Daymet incident solar radiation as the radiative channel DH (a physical replacement of unknown scale relative to the original 3-station DH); wind speed is included from gridMET in the 7-variable configuration. The median margin over damped persistence at 3–7 days is small (the win-rate carries the result), and a strong gradient-boosting baseline (LightGBM) is **at least as accurate as ThermoRoute**: significantly better at 1 day and statistically indistinguishable at 3–7 days once the seed budget is matched (per-seed protocol, §4.2). We therefore claim a robust improvement over the *physics* baselines and calibrated, transferable uncertainty — not a new state of the art over all learners. We report five seeds. The dynamic- κ thermal-memory modulation does **not** generalise: the flow-dependence seen on three stations vanishes on the large sample (κ rises with flow at 0 % of stations), and freezing κ ’s modulators does not worsen RMSE, so we retract the mechanism claim and keep only the router’s interpretable, horizon-dependent driver shares.

Input uncertainty. Our forcings are gridded reanalysis (Daymet, gridMET), which we treat as exact inputs — the very assumption that BATEA (Kavetski et al., 2006a, 2006b) warns against for rainfall–runoff, where forcing error can dominate and, if ignored, biases the calibrated parameters. Two points make this less acute here than in the rainfall–runoff setting BATEA addresses. First, water temperature is far smoother and more strongly autocorrelated than discharge, so short-horizon skill is carried by the persistent state (WTEMP history) rather than by the noisy meteorological forcing, limiting the leverage of input error. Second, we do not need the input-error model that BATEA requires and that it concedes is “poorly understood”: the conformal layer calibrates the predictive intervals *empirically* on held-out years, absorbing residual forcing error into the coverage rather than attributing it. The cost, in BATEA’s terms, is that we cannot separate forcing error from model error — a deliberate trade of process attribution for distribution-free coverage at scale. A fully operational system would additionally replace our reanalysis forcings with archived numerical weather-prediction forecasts, whose own error would degrade multi-day skill and could be handled either by a BATEA-style latent input model or by re-calibrating the conformal intervals on forecast-driven residuals; we leave this operational-forcing track to future work.

Equifinality. The near-identical validation loss across our five seeds (1.343–1.348) is a learned-model echo of the parameter equifinality that motivated GLUE Beven and Binley (1992): many weight configurations simulate the calibration data almost equally well. Where GLUE embraces this and propagates it into the prediction limits, our bounded physics prior instead *reduces* it — constraining the residual so the admissible solutions cluster near damped persistence — which is philosophically closer to BATEA’s add-structure stance than to GLUE’s accept-equifinality one, and is consistent with the low across-seed spread we observe.

6 Conclusions

Under a strict historical-information protocol, ThermoRoute matches per-station damped persistence where the system is near-deterministic (a reservoir cascade) and beats the physics baselines where forecast headroom exists (a 120-station large sample), transferring to unseen basins and providing near-nominally-calibrated warnings. We deliberately report two negative results — no point-accuracy gain on the cascade, and no generalisable flow-dependent thermal memory — to keep the claims honest. The contribution is a calibrated, transferable, interpretable forecaster whose advantage over the physics baselines is established on a large, diverse sample rather than a single site, with the explicit caveat that a strong gradient-boosting learner is at least as accurate across leads — better at 1 day and statistically on par at 3–7 days — so the claim is calibrated, transferable skill over the physics baselines, not a new state of the art over all learners.

Open Research

Data availability

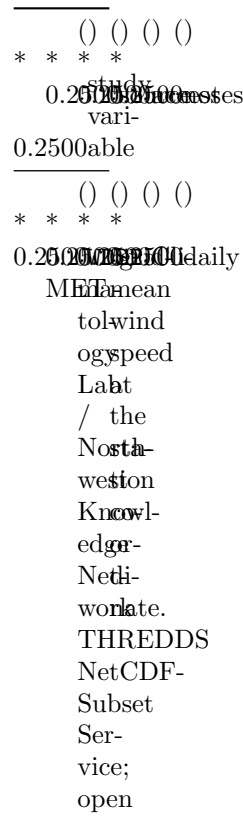
This study uses two datasets.

(1) Three-station reservoir cascade (case study). Daily records (2006-01-01 to 2020-12-31; 5 479 days per station) of water temperature, discharge, reservoir level, air temperature, precipitation, wind speed, mean relative humidity and a radiative index (DH) at three cascade stations (b1, s2, p3). These were provided for this study; the DH field’s data-dictionary definition is unconfirmed and we make no DH-specific physical claim. Two sentinel missing-codes (WDSP=999.9, PRCP=99.99; 0.1–0.3 % of records) are masked to NaN and imputed within the training fold.

(2) 120-station USGS large sample (main analysis). Assembled programmatically from three open sources, in the same schema, by `scripts/data_usgs/build_usgs_stations.py`:

[illegible]

[illegible]



120 stream gages across 35 U.S. states and 16 HUC2 regions ($\geq 55\%$ water-temperature and $\geq 70\%$ flow coverage over 2006–2020, plus a $\geq 80\%$ blind-test-window coverage gate — §2) span free-flowing and dam-regulated rivers; water-temperature coverage ranges 0.56–1.00 (median ≈ 0.88). The assembled main panel is `data_usgs/panel_usgs_100.parquet` (120 stations, seven model variables plus WLEVEL; the filename is historical and pre-dates the 120-station rebuild). Station metadata with USGS site numbers and coordinates is in `data_usgs/stations_meta_120v2.csv`, and every probed-but-rejected candidate is recorded in `data_usgs/rejected_sites_120v2.csv` (`outputs/reports/usgs_acquisition.md` documents the audit trail). The legacy 40-station pilot panels (`data_usgs/panel_usgs.parquet`, `data_usgs/panel_usgs_wind.parquet`) are retained only for the §S1 pilot comparison. All panels are included so the experiments reproduce without re-downloading; the raw per-site downloads can be regenerated by re-running the acquisition script. All three sources are public domain or open-licensed.

Code availability and reproducibility

All code, the fixed train/validation/calibration/blind-test boundaries, unit tests (leakage, splits, metrics, conformal, cross-model sample consistency), and one-command re-production (`scripts/run_all.sh`) are in this repository. Per-day predictions, model weights and logs are regenerable by the staged scripts (`scripts/01-14`); `scripts/14_manifest.py` writes a sha256 manifest of every artifact the manuscript’s numbers depend on (`outputs/manifest.json`), and `--check` verifies the recorded hashes so any drift between the manuscript and the artifacts is machine-detectable. Continuous integration runs the test suite under both the version-locked environment that produced the published numbers (`requirements-lock.txt`) and loose forward-compatibility floors. An adversarial internal review (multiple independent expert lenses) checked every headline claim against the result tables; all negative results and ablations are retained, and a finding-by-finding disposition is provided in `outputs/reports/review_r`

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